Lab 2 - H-Bridge with DC Motor

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Abstract

Motors are widely used in robotics, industrial machinery and automation, and many other industries. By understanding the schematic diagram of a specific motor givin in the lab, we build a circuit based on $ArduinoKona328$ and $chip - L293D$ to run the motor. We demonstrate adjusting speed, and then reverse its rotation direction. Additionally, since light-dependent registers (LDRs) can be used to change speed and direction, a light focused car motor design is presented in this lab report.

Introduction

This experiment aims on discovering the basic working principle of a simple motor. Then a theory of experiments would be An Arduino programmed circuit is built to run the motor.

- Set up motor to be running;
- Change motor to a certain speed;
- Run motor forward and reverse.

The motor is placed in a breadboard, and powered by a voltage up to 5V. Motor's speed is adjusted by changing voltage, with a
Give a brief description for the aims of the laboratory and a summary of the results collected. Please write this as constant prose in a way that a fellow student, who does not know anything about the experiment, can understand the approach.

Theory and Technical Background

Give a brief description of the background theory for the experiment being tackled. Please cite appropriate work. Feel free to add figures into this section, where each figure and each equation may reduce the number of words you use.

Method

Describe the technical procedures to collect the results. Add figures into this section, where each figure and each equation may reduce the number of words you use.

Results

Add all appropriate figures and analysis. No formal word count for this section, however please keep text as concise as possible.

Conclusions

Please give the conclusions from the characterisations including a summary of the results collected.